



Vibrio Cholera

SGL 7

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Clinical case

A 27-year-old man from an area reporting cholera presents with sudden profuse watery diarrhea and vomiting. He is afebrile, very thirsty, tachycardic, and hypotensive. The stool is pale and watery with flecks of mucus.

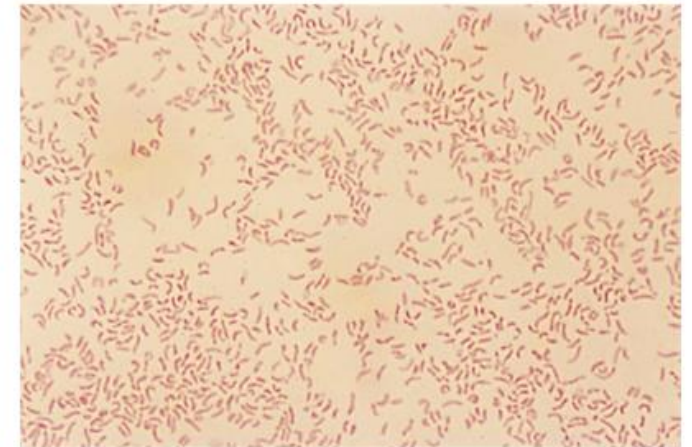
- What is the most likely diagnosis?
- What is the most urgent management step?
- What specimen would you collect?
- Which culture medium is classically used for isolation?

Vibrio

- Vibrios are **curved rods** with a **single polar flagellum**. They belong to the family *Vibrionaceae*.
- ***Vibrio cholerae***, the major pathogen in this genus, is the cause of cholera.
- *Vibrio* shares many **cultural and physiological** characteristics with members of the *Enterobacteriaceae*, a closely related family.
- Vibrios are **fermentative** and grow on **ordinary or selective** media containing bile at 37°C.
- They possess **unique O and H antigens** and membrane receptor antigens that provide some basis for classifying members of the family.



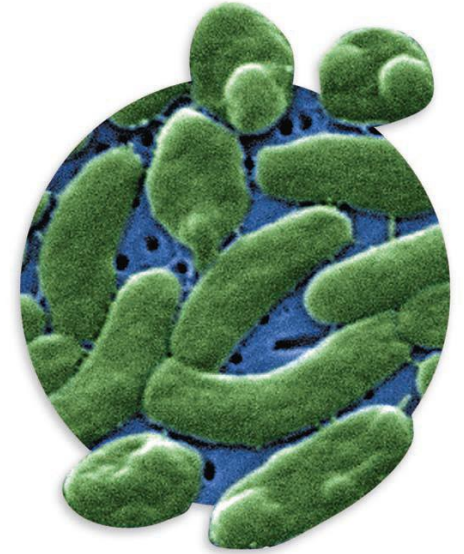
Comma-shaped bacillus with polar flagellum



Gram stain of *Vibrio* spp. A Gram stain of *Vibrio* spp. shows typical curved and straight, Gram negative bacilli

Vibrio cholerae

- Gram-negative, **curved or comma-shaped** bacillus.
- Motile by a **polar** flagellum.
- **Oxidase positive** and facultatively anaerobic.
- Grows well in **alkaline** conditions.
- **Toxigenic** strains colonize the small intestine and produce cholera toxin.
- Divided into serogroups according to the nature of its **O cell wall antigen**.



Vibrio cholerae
curved shape of this bacterium.

Serogroups and Classification

- More than **200 O-antigen** serogroups of *V. cholerae* have been described.
- Only toxigenic **O1** and **O139** strains have caused widespread epidemic cholera.
- **Non-O1** organisms either cause sporadic disease or are nonpathogens.
- **The O1 organisms have two biotypes:**
 - **Classic**
 - **El Tor** (responsible for the current global pandemic).
- **O1 serotypes: Ogawa, Inaba, and Hikojima.**
- O139 is now reported much less frequently and has been largely restricted to Asia.

(**Biotypes** are based on **differences in biochemical reactions**, whereas **serotypes** are based on **antigenic differences**.) These features are used to characterize isolates in epidemiologic investigations.

Transmission and Epidemiology

- The pattern of cholera transmission and the onset of epidemics are greatly influenced by the **season** of the year and the **climate**.
- **Cold, acidic, dry** environments **inhibit** the migration and survival of *Vibrio*, whereas **warm, monsoon, alkaline**, and saline conditions **favor** them.
- The bacteria **survive in water sources** for long periods of time.
- A person with normal gastric acidity may have to ingest as many as 10^{10} or more *V. cholerae* to become infected when the vehicle is water because the organisms are susceptible to acid.
- When the vehicle is food, as few as $10^2 - 10^4$ organisms are necessary because of the buffering capacity of food.



- Throughout history, cholera has triggered **seven pandemics**.
- Sadly, the year **2010** stands out for the reemergence of cholera in Haiti. After nearly a century without the disease, cholera bacteria were unintentionally introduced to the country by relief workers aiding after a catastrophic earthquake. Since then, cholera has led to more than 770,000 infections and claimed at least 10,000 lives.
- One of the world's preeminent microbiologists, **Rita Colwell**, has been studying *Vibrio cholerae* and safe drinking water for over 50 years, particularly in regions like Bangladesh where cholera outbreaks occur seasonally.
- She discovered that most cholera bacteria attach to plankton called copepods. By using folded sari cloth as a simple water filter, 99% of the bacteria could be removed, cutting cholera cases by 50% and greatly improving water safety.



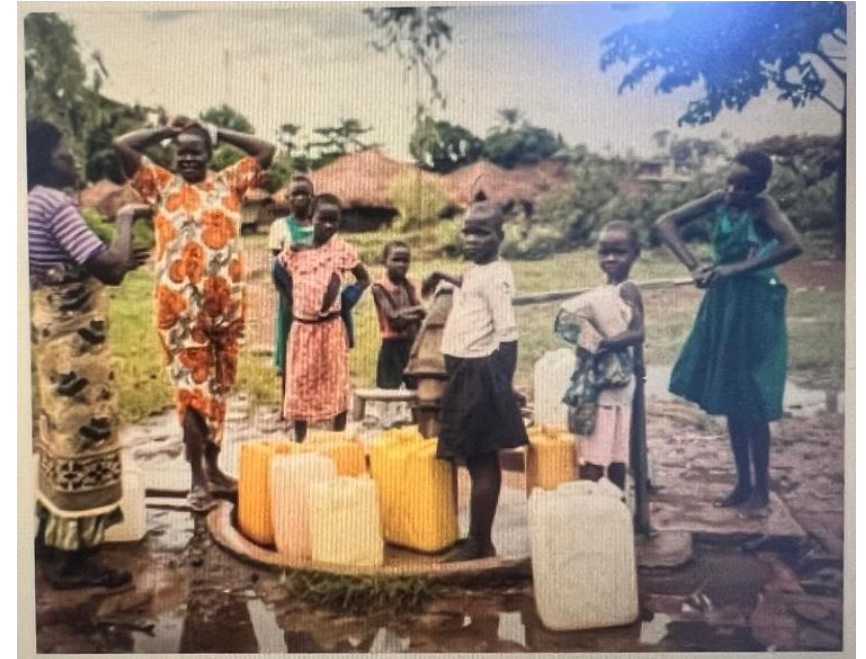
Rita Colwell



her sari solution

Who Is at Higher Risk?

- People without reliable access to safe water, sanitation, and hygiene.
- Populations affected by displacement, disasters, conflict, flooding, or crowded living conditions.
- Travelers, aid workers, and healthcare workers exposed in areas of active transmission.
- Reduced gastric acidity increases susceptibility to infection.



Signs and Symptoms

- After an incubation period of a **few hours to a few days**, symptoms begin abruptly with **vomiting**, followed by **copious watery feces**.
- Watery diarrhea in large volumes is the hallmark of cholera.
- There are no red blood cells or white blood cells in the stool. **Rice-water stool** is the term often applied to the nonbloody effluent.
- There is no abdominal pain, and subsequent symptoms are referable to the **marked dehydration**.
- The loss of fluid and electrolytes leads to **cardiac and renal failure**.
- **Acidosis and hypokalemia** also occur as a result of loss of bicarbonate and potassium in the stool.
- The **mortality rate** without treatment is 40%.



Cholera Cots. Cholera, caused by *V. cholerae*, causes such profuse diarrhea that special cots are used during outbreaks to keep suffering patients clean. Centers for Disease Control and Prevention

Classic Clinical Presentation

- Sudden onset of profuse, painless watery diarrhea.
- “Rice-water” stool: pale watery stool with flecks of mucus.
- Nausea and vomiting may occur, especially early in severe disease.
- Thirst and painful muscle cramps reflect volume and electrolyte loss.
- Severe cases are often afebrile; blood in stool suggests another diagnosis.

Note/ The key danger is not the number of stools — it is the speed and magnitude of fluid loss.

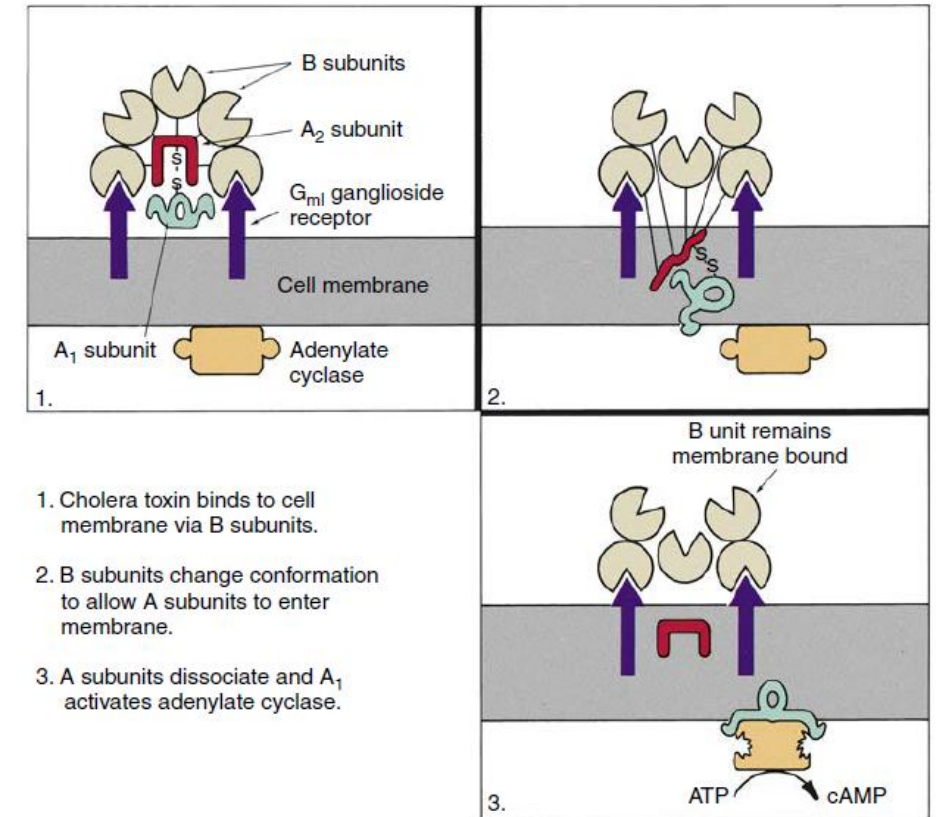


Rice-water stool

Pathogenesis and Virulence Factors

- After being ingested **with food or water**, *V. cholerae* travels through the stomach to the small intestine.
- At the junction of the duodenum and jejunum, the vibrios penetrate the mucous barrier using **their flagella**, adhere to the microvilli of the epithelial cells, and multiply there.
- The bacteria adhere to the intestinal mucosa of the small intestine, where they are **not invasive** but secrete **cholera toxin, an AB toxin**.
- Ganglioside GM1 serves as the mucosal receptor for subunit B, which promotes entry of subunit A into the cell.
- The **A subunit enters** the intestinal epithelial cells and adds an ADP-ribosyl group to the G protein coupled receptor, Gs α , which becomes locked in the activated state.

- This results in **constant stimulation** of host adenylyl cyclase and overproduction of cAMP. This leads to **hypersecretion** of water and chloride ions while **inhibiting** absorption of sodium ions.
- The infected person loses **massive quantities** of fluid and electrolytes, causing abdominal **muscle cramps**, **vomiting**, **fever**, and **diarrhea**.
- It was recently discovered that *V. cholerae* uses **quorum sensing** to regulate the precise expression of its virulence factors, making proteins used in this process potential targets for drug therapy.



The structure and action of cholera toxin

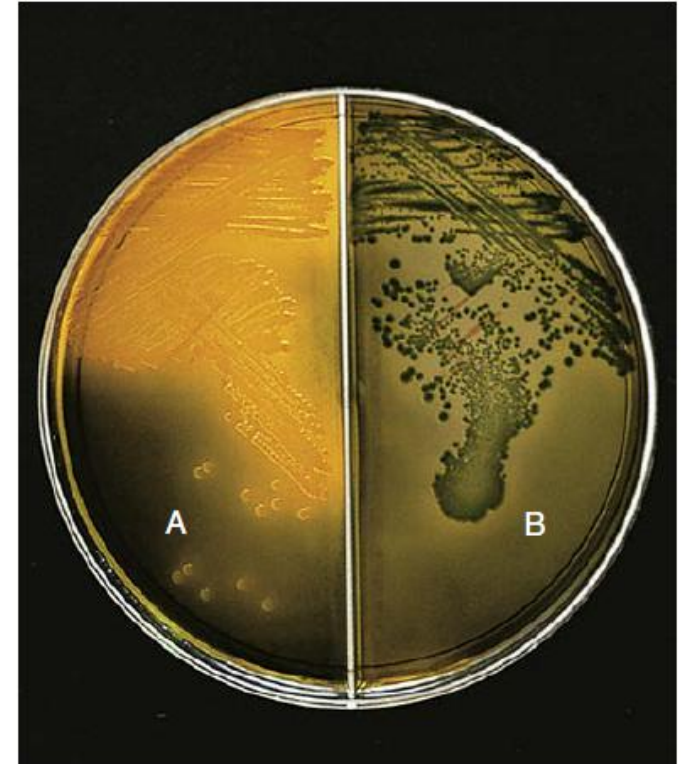
Laboratory Diagnosis

The approach to laboratory diagnosis depends on the situation. During an epidemic, a clinical judgment is made and there is little need for the laboratory.

- *V. cholerae* can be readily isolated and identified in the laboratory from stool samples.
- Direct dark-field microscopic observation reveals characteristic **curved cells with brisk, darting motility** as confirmatory evidence.
- The organism is **oxidase-positive**, which distinguishes it from members of the Enterobacteriaceae
- **Group-specific antisera** is supportive as well.



- a culture of the diarrhea stool containing *V. cholerae* will show colorless colonies on MacConkey's agar because lactose is fermented slowly.
- Thiosulphate–citrate–bile salts–sucrose (TCBS) agar is a selective medium widely used for the isolation of *V. cholerae*.
- On TSI agar, an acid slant and an acid butt without gas or H₂S are seen because the organism ferments sucrose
- Real-time PCR methods are now being employed to rapidly test food and water supplies for the presence of *V. cholerae* and related pathogenic species (*V. vulnificus* and *V. parahaemolyticus*).



Colonies of *Vibrio cholerae* (A) and *Vibrio parahaemolyticus* (B) on thiosulfate citrate bile salts sucrose agar.

Treatment

- Prompt, adequate **replacement of water and electrolytes**, either orally or intravenously. **Glucose** is added to the solution to enhance the uptake of water and electrolytes.
- Antibiotics are adjunctive therapy for severe cholera and selected high risk patients (not routine therapy for every mild case). They can reduce the duration of diarrhea, stool volume, and bacterial shedding.
- **Doxycycline and azithromycin** are commonly used options; choice should follow local susceptibility patterns and patient factors.
- Children 6 months to 5 years with suspected cholera **should receive zinc** when available. Zinc can reduce the severity and duration of diarrheal illness.
- Pay close attention to **hypoglycemia, dehydration**, and ongoing stool losses in children.

Prevention

- Public health measures that ensure a clean water and food supply.
- An oral, live attenuated vaccine called Vaxchora is available in the United States for travellers to areas where cholera caused by serogroup O1 is endemic. Other oral vaccines containing killed organisms are available in countries where cholera epidemics occur.
- The use of tetracycline for prevention is effective in close contacts but does not prevent the spread of a major epidemic.
- Prompt detection of carriers is important in limiting outbreaks.

Non-Cholera Vibrio Species

- In the United States, it is more common to experience infection with a non-cholera species of *Vibrio* than a *V. cholerae* infection.
- These infections are called **vibrioses**, as opposed to cholera. Two species are most prominent :
 - ***V. vulnificus***
 - ***V. parahaemolyticus***.
- Infection can be from exposure to **seawater** but more often is associated with **eating contaminated shellfish**.
- It is a relatively rare infection but on the increase. Scientists suspect that the increase is due to three factors:
 - (1) Increased consumption of raw oysters;
 - (2) Increased awareness, meaning that more people are diagnosed;
 - (3) A warming climate, causing a wider habitat for these bacteria in bodies of water.
- In people who are immunocompromised, the infections can be fatal, especially in the case of *V. vulnificus*. These infections, along with those by *V. cholerae*, are nationally notifiable diseases.

Case Study

An 18-year-old woman, previously healthy, was hospitalized 7 days after developing severe watery diarrhea. She had recently returned to the U.S. from Ecuador. About 8 days prior to admission, she ate in a restaurant. All who were with her ate fish, as she did, but she was the only one who ate fresh vegetables. Two days later she developed profuse, watery "rice water" diarrhea, followed by vomiting. She had no fever or chills. On admission, she was dehydrated, hypotensive (BP 88/50), with dry skin, sunken eyes, and dry mouth, indicating significant fluid loss. The remainder of the physical examination was within normal limits. Wright stain of stool revealed no leukocytes. The stool was negative for occult blood. Culture of stool yielded an organism. The most likely organism to cause this illness, is:

- A. *Escherichia coli* O157:H7.
- B. *Salmonella typhi*.
- C. *Clostridium difficile*.
- D. *Shigella sonnei*.
- E. *Vibrio cholerae*.

- The correct answer is
- **E** (*Vibrio cholerae*). It causes disease by elaborating a toxin that greatly increases the influx of ions and water into the lumen of the intestine. As a secretory, noninflammatory diarrhea, it is not characterized by fecal leukocytes or blood.
- **A** (*Escherichia coli* O157:H7) causes hemorrhagic colitis, with copious stool as this patient had, but patients with this infection usually have very bloody stools. The reservoir is cattle, not fish or vegetables.
- **B** (*Salmonella typhi*) causes enteric fever (this patient was never febrile) and, when there is diarrhea, it is often preceded by constipation. It is an inflammatory process, so that there are fecal leukocytes.
- **C** (*Clostridium difficile*) usually, but not always, causes disease in relation to health care, especially during or after administration of antibiotics. The associated diarrhea is intensely inflammatory with very numerous fecal leukocytes.
- **D** (*Shigella sonnei*) is also a cause of extremely inflammatory diarrhea, with so many fecal leukocytes that pus may be visible in stool without magnification.

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